

AI-Powered Advertising Strategy Optimization and Click Prediction for Improved Digital Marketing

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Abstract— In contemporary times, traditional marketing methods have transformed, giving way to online marketing approaches that necessitate internet connectivity. Online marketing strategies offer numerous benefits, particularly in terms of cost-effectiveness and expeditious dissemination of information to the general populace. Nevertheless, one of the primary issues faced by firms in the realm of Internet marketing pertains to the identification of appropriate target consumers. This is because if organizations mistakenly target consumers not aligned with their products or services, it can result in ineffective marketing efforts and waste of resources. This research thoroughly examines ad click prediction and classification through a machine learning (ML)-based novel approach. Predictive models are developed by utilizing past ad interaction data, encompassing user demographics, ad features, and contextual information. The proposed method identifies the important factors driving ad-click behavior more easily using feature engineering approaches to extract pertinent patterns and correlations from the data by using the Pearson correlation method. Among all the implemented algorithms, the highest accuracy of 97.5% and AUC of 0.996 on test data is achieved by the logistic regression classifier. This research adds to the body of knowledge that will help advertisers pinpoint their ideal target demographic, fine-tune their ad placement techniques, and make the most of the resources they have at their disposal.

Keywords—Machine learning, SVM, Neural networks, CTR, click, advertisement.

I. INTRODUCTION

The utilization of online advertising possesses significant potential to enhance the income of businesses. The click-through rate (CTR) is an essential metric for assessing the efficacy of Internet advertising campaigns. The metric quantifies the proportion between the aggregate clicks received and the overall number of engagements. Hence, using click prediction techniques has become imperative for enterprises as they strive to enhance the CTR of their online marketing efforts by accurately forecasting user clicks [1]. Hence, predicting user engagement with advertisements, namely the likelihood of a click, has garnered significant interest among researchers.

Display advertising refers to online advertising that involves the placement of banner adverts, including various media types, such as text, images, videos, audio, and motion, within predetermined sections of a website or application. It is projected that global search advertising expenditure will reach a total of 164.6 billion U.S. dollars in the year 2022. The figure is expected to rise to almost \$190.5 billion by 2024. In 2021, display advertising emerged as the dominant form of advertising, with a significant ad spend share of 244 billion U.S. dollars. [2]. The significant surge in online advertising has prompted a need for a more accurate

evaluation of the CTR to ascertain the cost-effectiveness of investing in digital advertising. A greater CTR indicates that the advertisement has achieved success in attracting attention.

The Ad click can be influenced by various factors, including the banner's placement, the display's size, and the specific website on which the advertisement is being shown [3]. Furthermore, several online advertising systems, such as Google Ads and Facebook Ads, employ an ad ranking method that relies on multiplying the CTR and the bid. The bid represents the monetary value businesses are prepared to allocate for a particular action, specifically clicking an advertisement. By knowing the anticipated CTR by click prediction before launching an advertising campaign, firms may optimize their revenue generation while ensuring a satisfactory user experience [4].

Applying machine learning (ML) algorithms in predicting ad clicks has gained significant attention in current research [5]. This interest is fueled by the diverse possibilities that ML presents for revolutionizing the field of digital advertising. The use of ML methods in this field is driven by the need to improve targeting accuracy, maximize cost-efficiency, and enhance the entire return on investment for firms operating in the complex online marketing landscape.

Using advanced machine learning models, advertisers can identify intricate trends in user behavior, facilitating the development of highly precise advertising tactics that effectively resonate with distinct audience segments. Utilizing these sophisticated algorithms enables the dissemination of customized and contextually relevant adverts, promoting increased user involvement and enhanced conversion rates. The dual advantage of this approach lies in its ability to optimize the allocation of resources for advertisers, leading to improved budget distribution efficiency. Additionally, it enhances the consumer's experience by providing advertisements that closely align with individual preferences, thereby fostering a more immersive and personalized digital interaction [6]. In addition, the knowledge obtained from thoroughly examining user data enables advertisers to make informed choices based on data, guaranteeing the ongoing improvement and enhancement of advertising strategies to achieve the highest effectiveness and efficiency in the highly dynamic digital business.

Other than digital marketing, ML has significantly impacted various areas, leading to transformative changes in operations and promoting innovation across a wide range of businesses. ML plays a crucial role in the healthcare sector by facilitating disease diagnosis by examining intricate medical data [7]. As a result, ML has effectively hastened

the advancement of novel treatments. ML plays a crucial role in the financial industry by being an essential tool for detecting fraudulent actions [8]. It achieves this by utilizing sophisticated algorithms to discover irregularities and suspicious behaviors within extensive transactional data. Furthermore, the significance of ML in the path trajectory [9] is significant, along with other applications like indexing [10], databases [11], and camera focusing [12].

This research aims to present a novel approach for predicting advertisement clicks utilizing machine learning (ML) algorithms in response to the increasing use of ML in diverse fields and its growing prominence in digital marketing. The ML-based strategy in this study is proposed to include several steps, such as pre-processing the dataset to handle missing values, evaluating the data to identify the most relevant features, and scaling the numerical values. The pre-processed dataset is utilized in ML techniques such as Logistic Regression (LR), Support Vector Machines (SVM), Decision Tree (DT), and Random Forest (RF). The fundamental performance classification measures are used to evaluate the effectiveness of all models.

II. LITERATURE REVIEW

An increasing number of studies using various machine learning (ML) approaches have contributed to ad click prediction in online marketing. The literature comprehensively examines predictive modeling techniques, ranging from fundamental logistic regression models to sophisticated deep learning architectures. These techniques are employed to understand user behavior and improve strategies for promoting ad interaction.

The research by Dani and Ginting [13] develops a consumer ad click prediction model utilizing an ML approach that demonstrates increased sophistication in accurately estimating the likelihood of a click. The two classification algorithms, LR and k-nearest neighbors (k-NN) classifier, are used. The experimental results demonstrate that the logistic regression technique exhibits superior performance by achieving an accuracy of 91%.

Similarly, Shah and Nasnodkar [14] present new user experience measures for predicting CTR in online advertising. The dataset undergoes preprocessing techniques to address imbalances. The original and the balanced datasets are used for training and testing LR, DT, and RF. The study found significant associations between CTR and personalization, with a positive correlation of 0.47 and a negative correlation of -0.38. Incorporating personalization yielded a notable effect, as evidenced by a forecast of CTR with a score of 0.25. The findings indicated that both DT and RF had shown strong predictive abilities, particularly when trained on balanced class distribution datasets. These models achieved the maximum level of accuracy, reaching 89%, when evaluated on test data.

Zhang [15] also explores the conceptualization of advertising and its practical implementation by examining ten distinct types of advertising. A set of features was extracted. In addition, the paper first employs SVM, a basic Bayesian model, and a decision tree algorithm. Using both tree models and neural network models are employed to forecast the click-through rate and analyze their performance. The experimental findings demonstrate that the SVM model approach employed in this study achieves the highest accuracy of 93.24 % on test data.

As mentioned above, Alali et al. [16] suggested establishing a model that produces precise outcomes while requiring minimal computational resources. In this study, four classification algorithms were employed, specifically KNN, LR, RF, and Extreme Gradient Boosting (XGBoost). The research was conducted using the dataset from the CRP Competition. The data in question is a collection of click-through information that has been organized in chronological order. This data was gathered for ten days. The experimental findings indicate that the XGBoost algorithm achieved an Area Under the Curve (AUC) value of 0.76 when a restricted set of features was utilized.

Likewise, Sahllal and Souidi [17] assess the effectiveness of 19 class imbalance approaches when applied to a native ads dataset. Four standard classifiers are utilized to discover the absolute effective imbalance methods. The dataset utilized in this study comprises genuine traffic data obtained from Finland, specifically derived from ReadPeak. The employed resampling strategies encompass seven ways of under-sampling, four techniques of oversampling, four hybrid sampling approaches, and four ensemble systems. The findings indicate that the utilization of class imbalance learning techniques can yield notable enhancements in the model's classification performance. This is evidenced by a substantial increase in accuracy, up to 20%. Using oversampling techniques demonstrates enhanced stability, whereas combining under-sampling methods with the RF algorithm has an accuracy of 66.66%.

Like the above approaches, Asad et al. [18] introduce an AI-based framework for identifying factors that contribute to the enhanced viewability of advertisements. Feature selection techniques are employed to retrieve important features from the dataset. After that, clustering ensures that the examples have the right of class labels. The training uses five distinct classifiers: SVM, k-NN, RF, NN, and Gradient Regression Boosting Trees.

On the other hand, Li et al. [19] introduce InterHAt, a novel approach that utilizes a Transformer model with a multi-head self-attention mechanism for feature learning. Furthermore, the utilization of hierarchical attention layers is employed to predict CTR while offering interpretable insights into the results of the predictions. The InterHAt model successfully measures high-order interaction between features by utilizing an attentional aggregation technique that is both efficient and computationally lightweight. The efficacy and efficiency of InterHAt are demonstrated through comprehensive experimentation on four publicly available real datasets and one synthetic dataset. The results indicate that InterHAt achieves the greatest AUC value of 0.9696.

In contrast to the above approaches, Ouyang et al. [20] introduce the Mixed Interest Network (MiNet) to optimize news data utilization for predicting the CTR of advertisements. MiNet encompasses three distinct forms of user interest: lasting interest that spans multiple domains, transitory interest that originates from a given domain, and transitory interest specifically focused on the target domain. The MiNet system has two attention processes: item-level, which constantly retrieves pertinent information from selected news or advertisements, and interest-level, which effectively integrates diverse interested expressions. MiNet has been implemented in UC Toutiao, resulting in a significant improvement in online click-through rate.

Alternatively, Jiang et al. [21] introduce a new method for estimating CTRs using a multi-view feature transfer (MFT) approach. MFT categorizes data into common characteristics and selected features, then integrates significant feature vectors from a Laplacian matrix with conventional characteristics to create perspective clusters. This innovative approach is particularly useful for CTR data prediction. The MFT model has shown promising results in trials on various datasets and metrics, demonstrating its ability to leverage transfer relationships among attributes in advertising datasets. It outperforms other methods for predicting advertising click-through rates.

The literature review reveals a dearth of studies on Ad click prediction. A few researchers have employed machine learning techniques such as SVM, RF, ANN, k-NN, and others to classify Ad clicks according to their attributes. The utilization of standard datasets is not commonly observed in these research studies. In addition, certain research endeavors incorporate deep learning (DL) techniques, such as transformer-based models, while novel mathematical frameworks are also being devised. Hence, it is imperative to investigate ML techniques in conjunction with exploratory data analysis to identify the optimal features inside the database that exhibit the highest correlation with ad clicks. This study employs ML techniques to present a new approach for predicting and classifying ad clicks.

III. PROPOSED METHODOLOGY

The proposed methodology uses ML techniques to predict the Ad click. The flow of the proposed approach is described in Fig 1.

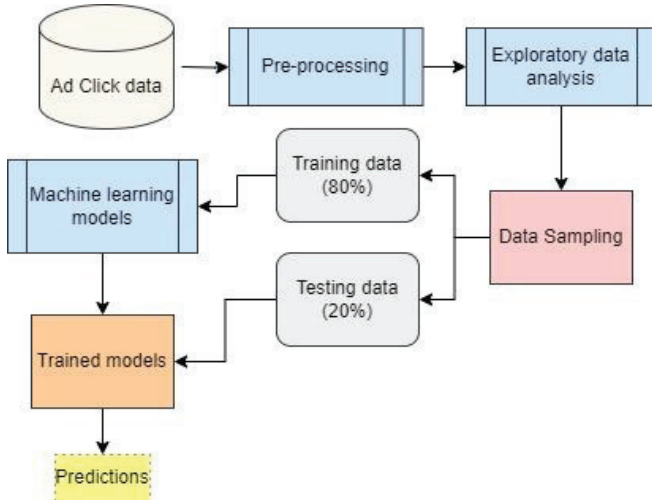


Fig. 1. Flow of the proposed approach.

A. Dataset Description

The Kaggle publicly available dataset named “Ad Click data” [22] is used for experimentation. The dataset consists of 1000 instances and ten variables. Among them, 9 are features, and one is the target variable, i.e., ‘Clicked on Ad.’ The features in the dataset include Daily Time Spent on the Site, Age, Area Income, Daily Internet Usage, Ad Topic Line, City, Male, Country, and Timestamp.

B. Data pre-processing

The dataset is subjected to analysis to identify and address any instances of missing values. The dataset has missing values in many variables. Specifically, there are 13

missing values in the variable ‘Daily Time Spent on Site,’ 13 missing values in ‘Area Income,’ 11 missing values in ‘Daily Internet Usage,’ 1 missing value in ‘City,’ 3 missing values in ‘Male,’ and 9 missing values in ‘Country.’ The missing variables about Daily Time Spent on Site, Area Income, and Daily Internet Usage have been imputed and substituted with the mean values. Although there are a limited number of missing values for the variables City, Male, and Country, they have been disregarded in the analysis.

C. Exploratory data analysis

The dataset is analyzed to find the relationship between the features and the target variables. The Pearson correlation is performed to understand the relation between the features and the target variable. The Pearson correlation coefficient is a statistical metric measuring a linear association's magnitude and direction between two continuous variables. The utilization of this method is prevalent in evaluating the extent of linear correlation between two variables. The Pearson correlation coefficient possesses a range of values between -1 and 1.

- A score of 1 signifies a flawless linear positive correlation, implying that as one variable experiences an increase, the other variable also undergoes a corresponding increase.
- A value of -1 denotes a precise negative linear correlation, implying that when one variable experiences an increase, the other variable experiences a proportional reduction.
- A zero value denotes the absence of a linear association between the variables.

D. Machine learning models

The training set (80%) is provided as an input to the ML models like LR, DT, RF, and SVM with linear kernel. The optimized hyperparameters are given in Table I.

TABLE I. OPTIMIZED HYPERPARAMETERS FOR ALL THE MODELS

Models	Hyperparameter	Value
LR	Penalty	12
	C	0.1
	Max_iter	100
	tol	1e-5
DT/ RF	criterion	Gini
	splitter	Best
	Max_depth	None
	Min_samples_split	2
	Min_samples_leaf	1
RF	n_estimators	100
SVM	kernel	Linear
	tol	0.001

1) Logistic Regression

The method used is to assess the probability of a binary result. The name "logistic" pertains to the logistic function employed within this regression model to convert a linear combination of input variables into an estimate of likelihood.

It is commonly employed when the target variable is binary or categorical, but the input variables can assume any

form. The objective is to construct a statistical model that estimates the likelihood of a specific category or result. The logistic function, alternatively referred to as the sigmoid function, transforms a real-valued variable into a range between 0 and 1, allowing for its interpretation as a probability. The mathematical expression for logistic regression is given by Eq. 1 [23].

$$P(Y = 1|X) = \frac{1}{1+e^{(\beta_0+\beta_1X_1+\beta_2X_2+\dots+\beta_nX_n)}} \quad (1)$$

Where,

$P(Y = 1|X)$ = The likelihood that the dependent variable has a value of 1 given the parameters of the independent variables.

$\beta_0, \beta_1, \dots, \beta_n$ = Coefficients

$X_1, X_2, \dots, X_n$ = Features or independent variables.

2) Decision Tree

Decision trees (DT) [24] are useful for investigating and elucidating data, with each core node representing a test conducted on a specific attribute. The branches indicate possible outcomes, while the leaf nodes represent the class label or anticipated value. Constructing a DT involves iterative data division through recursive partitioning based on the values of several attributes. The algorithm chooses the attribute that divides the data into subsets with the highest level of purity. However, the DT technique is susceptible to overfitting, especially when it becomes too complex and incorporates irrelevant information. To mitigate this, strategies include setting a minimum threshold for sample distribution, imposing a maximum depth limit, and conducting pruning operations after construction.

3) Random Forest

The random forest [23] is the ensemble learning technique that incorporates randomization in two aspects: selecting sample data to train each DT and selecting attributes for dividing each node. The random forest algorithm operates as follows:

Step 1: The process involves randomly selecting "k" features from a pool of "m" total features.

Step 2: A decision tree can be constructed by utilizing the chosen features, typically by employing a subset of the available data.

To generate several decision trees, it is necessary to repeat the process outlined in steps 1 and 2.

In classification tasks, each tree contributes its "votes" towards a specific class, and the class that receives the highest number of votes is considered the prediction made by the model. In regression problems, the technique computes the mean of the outputs generated by all trees. RF can mitigate overfitting and enhance the model's overall performance and resilience using random features and bootstrapping techniques. Furthermore, these methods offer a means of quantifying the relevance of characteristics, aiding in comprehending which features exert the most substantial influence on the predictions made by the model.

4) SVM

The primary concept underlying SVM [23] is the identification of an ideal hyperplane within the feature space. The hyperplane efficiently partitions the classes, enhancing the separation between the closest data points belonging to different classes. The selection of the optimal hyperplane in SVM relies heavily on identifying support vectors, which are

data points that exhibit the closest distance to the hyperplane. These support vectors play a crucial role in the overall accuracy and effectiveness of the hyperplane. It has demonstrated efficacy in managing datasets with many dimensions, as they can capture intricate associations among various variables. Moreover, these models exhibit resilience against overfitting, particularly in scenarios where the margin is optimized.

When employing the linear kernel in SVM, the decision function is expressed as a linear amalgamation of the input features. The determination of the decision boundary is contingent upon the support vectors, defined as the data points that exhibit the closest proximity to the hyperplane. The linear kernel function demonstrates computational efficiency and exhibits strong performance when used to large-scale datasets characterized by a high dimensionality of features.

E. Performance metrics

The standard performance metrics used for evaluating the performance of the models are Confusion metrics, accuracy, recall, precision, F1 score, and Receiver Operating characteristics- Area under curve (ROC-AUC).

Confusion matrices are given in Fig. 2.

		Actual	
		Actual Yes	Actual No
Predicted	Yes	True Positive (TP)	False Negative (FN)
	No	False Positive (FP)	True Negative (TN)

Fig. 2. Confusion matrix.

All the other metrics are dependent on the Confusion matrix terms. As depicted in the Fig 2. TP is several positive occurrences where a prediction was accurate; TN is accurately anticipated negative instances; FP is incorrectly anticipated positive cases; and FN is inaccurately projected several negative events.

The formulas for precision, recall, F1 score, and accuracy [7] are given in Eq. 2-5, respectively.

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

$$F1 \text{ score} = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4)$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

The ROC curve is a graph that shows how the sensitivity and specificity change in response to the level of discrimination threshold. The AUC is a numerical metric that captures the effectiveness of a classifier in a single scalar value, providing a concise overview of its performance.

IV. RESULTS AND DISCUSSION

Python packages such as Sklearn, pandas, and Numpy are commonly used for research on the Google Colab platform. To find the important features, the correlation is performed, and its matrix is given in Fig.3. The highest positive correlation is between the 'Daily Time Spent on the

Site' and 'Daily Internet usage,' with a value of +0.52. However, the highest negative correlation is between 'Clicked on Ads' and 'Daily Internet usage' with a value of -0.79. The most important variables that affect the target variables 'Click on Ad' are Daily internet usage, Daily time spent on site having a negative correlation of -0.75, Age with a positive relation of 0.49, and Area Income having a negative association of -0.48 and Male. Thus, these features are only considered for training the ML models.



Fig. 3. Correlation matrix

TABLE II. COMPARISON OF ALL PROPOSED METHOD ALGORITHMS ON TEST DATA

Models	Accuracy	Precision	Recall	F1 score	AUC
LR	97.5 %	97.9 %	96.8 %	97.4 %	0.996
DT	94.0 %	92.8 %	94.7 %	93.8 %	0.940
RF	95.0 %	94.7 %	94.7 %	94.7 %	0.995
SVM	96.5 %	95.8 %	96.8 %	96.3 %	0.996

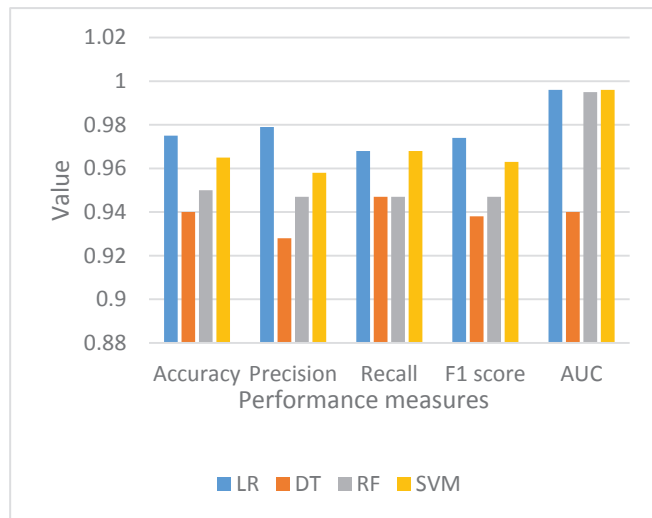


Fig. 4. Performance comparison of all the models

The outcomes of the tests done on each model are presented in Table II. According to the results presented in Table II, it is evident that the performance of the LR model surpasses that of the competing models by a substantial margin. LR achieves an AUC of 99.6% and an accuracy rate of 97.5%. The AUC of the SVM is equivalent to that of the LR model. The level of accuracy is 96.5%. The performance of the DT and RF models is comparatively worse when compared to the SVM and LR models.

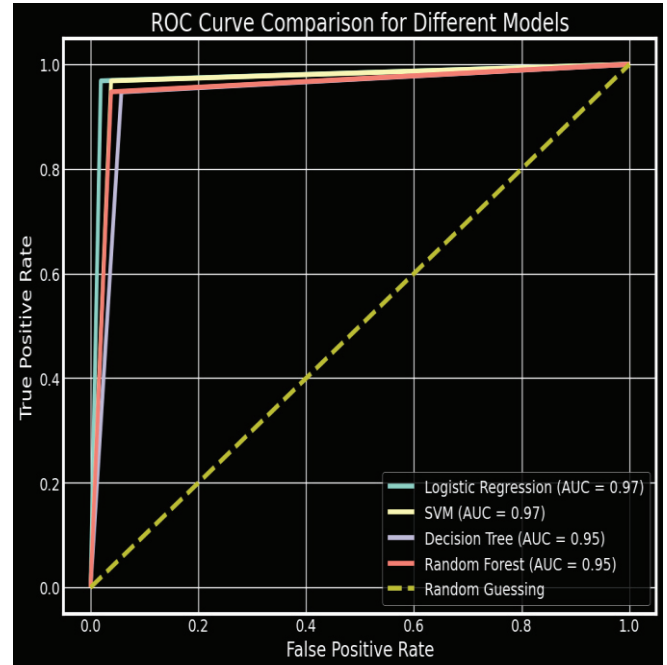


Fig. 5. ROC Curves of all the models

Fig 3 presents a visual representation of the performance metrics for all the models when evaluated on the test dataset. Fig 4 depicts the ROC curves corresponding to the various models that underwent training. Based on the data depicted in Fig 4, it is evident that the LR model, represented by the green graph, and the SVM model, represented by the yellow graph, exhibit a greater closeness to the point (0,1). The close relationship between these models implies that they demonstrate a higher level of performance in comparison to the other models.

V. CONCLUSION

The precise anticipation and categorization of ad clicks have grown crucial in optimizing advertising campaigns and maximizing returns due to the rapid expansion of Internet advertising. If there is a lack of interest in purchasing the product, the expenses associated with advertising will be elevated. Therefore, firms must own a click prediction system to accurately determine the appropriate course of action. The focus of attention is on the consumers who are being targeted. Various businesses and search engines heavily depend on modeling techniques to precisely forecast ad clicks. This research utilizes the ML-based approach for predicting the add clicks using the four ML algorithms LR, DT, RF, and SVM. Among all the models, the LR outperforms by achieving an AUC of 0.996 and an accuracy of 97.5%. The results emphasize the necessity of utilizing cutting-edge ML techniques in understanding complicated customer habits and choices, which pave the way for distributing targeted and pertinent adverts. Moreover, the

suggested methodology improves the effectiveness of Internet advertising and cultivates a more interactive and tailored user experience, ultimately promoting sustained client engagement and loyalty. The proposed methodology has the potential for future extension to enable real-time implementation and incorporate textual data for analysis.

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